RobotAI, Version 0919, **Last updated 11th October 2021**

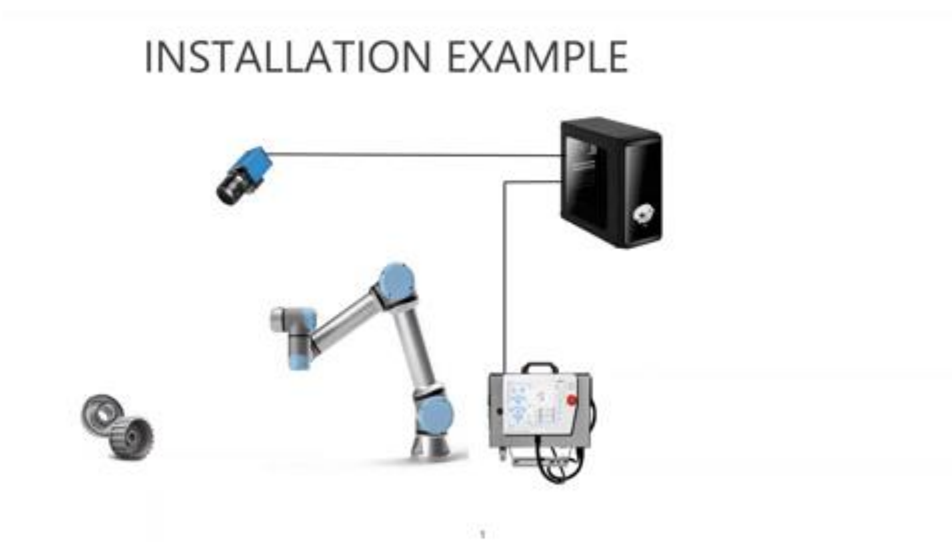
Pose6D software (CPU version) user guide

Preliminaries

Hardware and Software requirements:

1. The software is for Windows 10 OS.
2. The computer should have at least 8GB memory.
3. No GPU is needed. However, it will work slower than the GPU-version of the software.
4. In some cases, computers may require CUDA installation.
   Please see the troubleshooting page.
5. Installation of camera specific drivers:
   a. USB is built in and included for Windows 10.
   b. Other cameras require customization (contact RobotAI)

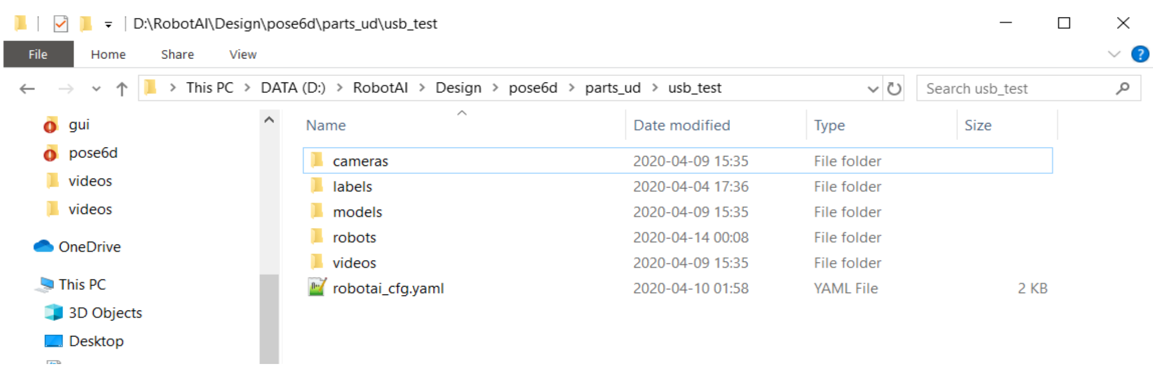
Figure below outlines the possible connection between camera, RobotAI computer and Robot controller.



IMPORTANT: In your setup and camera selection, size of the object should be at least 250 x250 pixels to receive accurate results.

Software Installation

1. Download the following file **Pose6D-XXXX-cpu.exe** to your computer folder. XXXX stands for a specific version.
   (say '**C:\RobotAI\SW**'. Substitute this string with your local computer location).
2. Create a directory for your objects/parts (say **C:\RobotAI\Parts**).
3. Download file usb_test.zip.
4. Unzip into the **C:\RobotAI\Parts** directory. You should have **usb_test** folder with following structure:

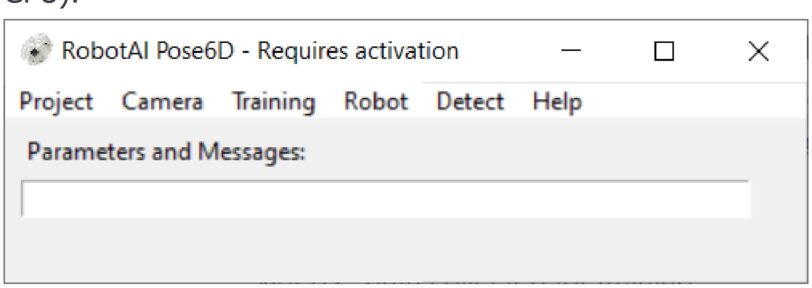


cameras - contains different camera related interface and parameters.
labels - labeled data used for model training.
models - contains run time models.
robots - robot interfaces and parameters.
videos - data collected for training.

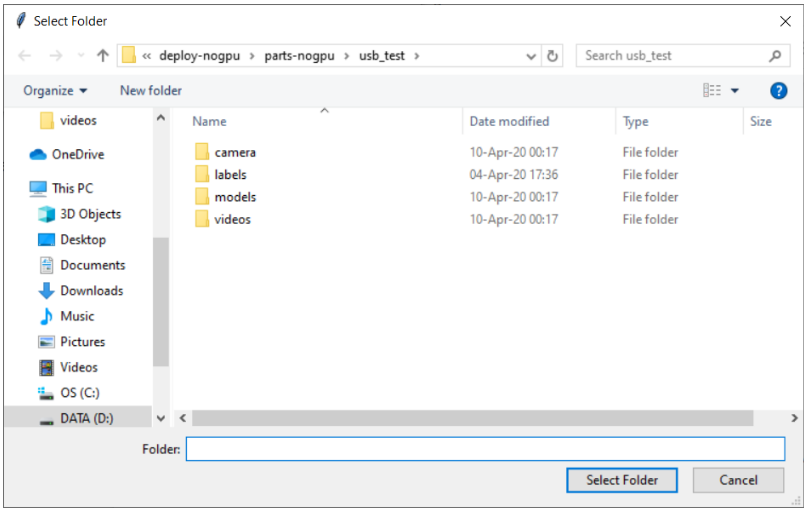
Software Verification

Recorded USB Data

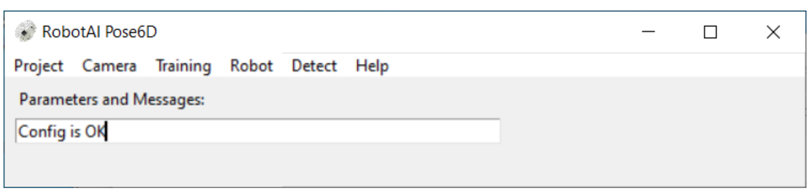
1. Run the software by double clicking on **C:\RobotAI\SW\Pode6D-XXXX-cpu.exe**. You should see the following window (it may take time to load since it is not using GPU):



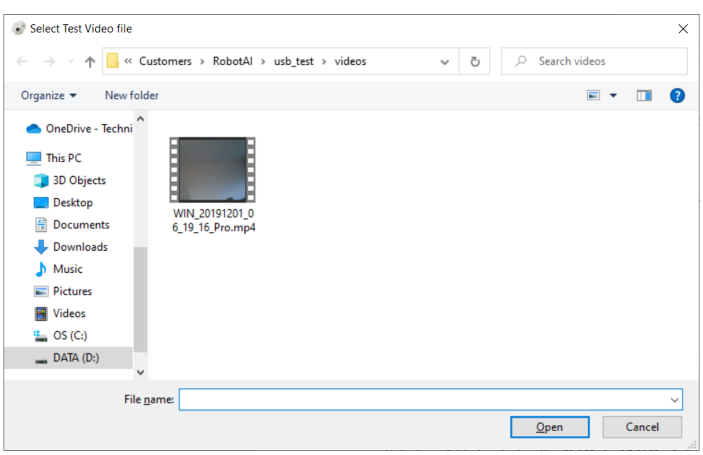
2. Now we will test the software. Go to **Menu bar> Project> Select Object Directory**. A window will open where you will need to select the working project directory. Select '**usb_test**':



3. Select **Menu bar>Project>Check Config File**. The window remains white and shows Config is OK (see below):



4. Select **Menu bar>Detect>Run Video from File** command. The following window opens.



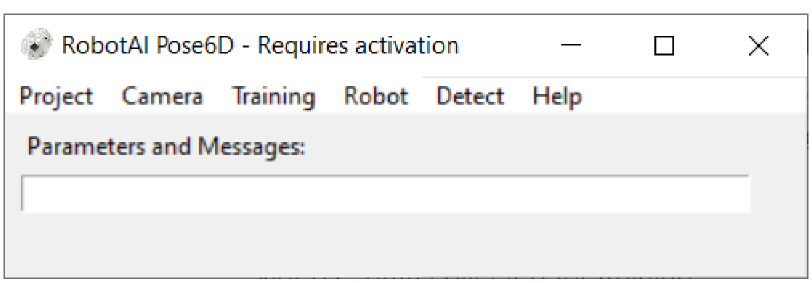
Select mp4 file located in the **C:\RobotAI\Parts\usb_test\video** directory. You should see a window with USB movie (see below).

Press '**q**' to stop. If the video runs fast - the installation is successful.

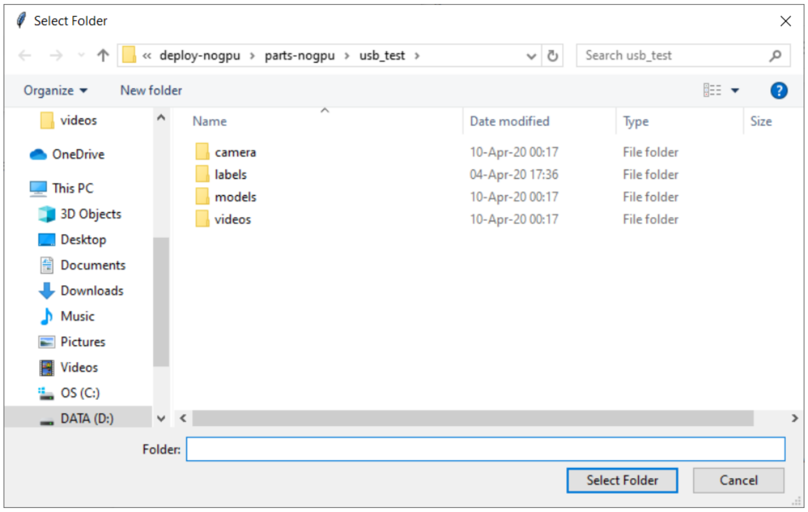
Your Own USB Data

This test will show USB detection on your Laptop or USB camera. Please note that these results are less reliable since we do not know your specific camera type and your light conditions. Please compensate for the background artifacts by adding a white paper in the background.

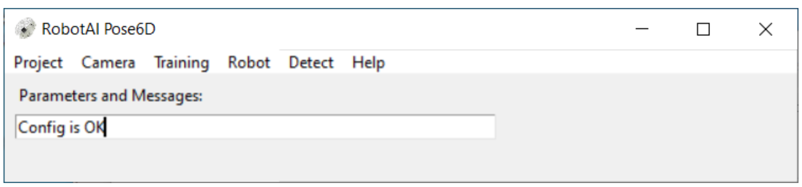
1. Run the software by double clicking on **C:\RobotAI\SW\Pode6D-XXXX-cpu.exe**. You should see the following window (it may take time to load):



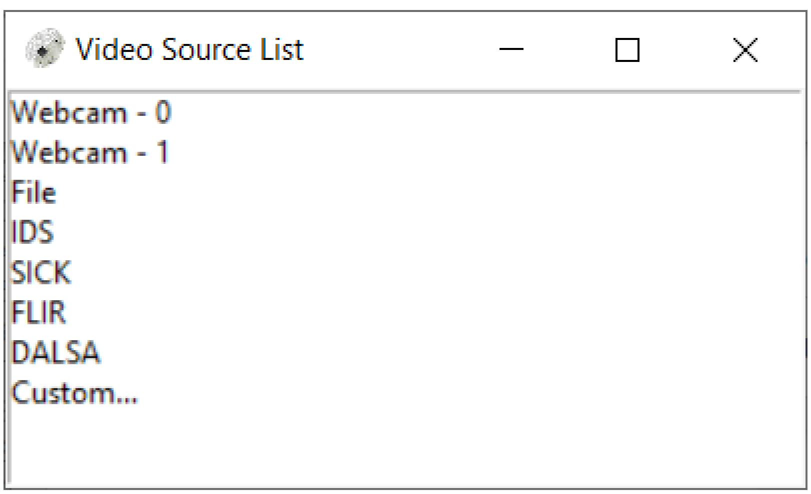
2. Now, we will test the software. Go to **Menu bar> Project> Select Object Directory**. A window will open where you need to select the working project directory. Select '**usb_test**'.



3. Select **Menu bar>Project>Check Config File**. The following window should pop up and remain white with message "Config is OK".

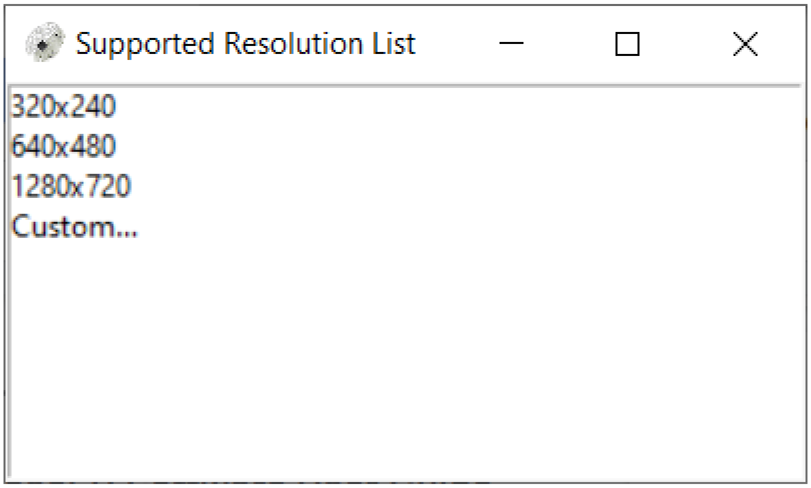


4. Go to **Menu bar > Camera>Select from List**.



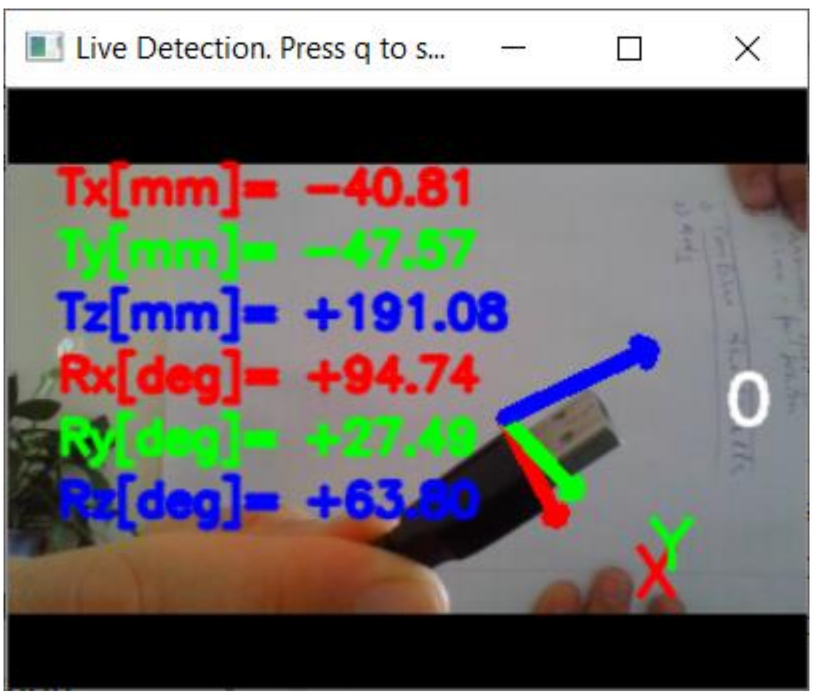
Select Webcam - 0 or 1 depends on your computer settings.

5. Next go to **Menu bar>Camera>Connect**. The camera is now activated.
6. Then go to **Menu bar>Camera>Configure > Resolution**.



Select 320x240. If your camera does not support this resolution, choose a different one or "Custom". Before selecting Custom resolution write in the command line something like this: 320x240 - the required resolution. Resolutions different from 320x240 make the detection less reliable.

7. Select **Menu bar> Detect > Run Video from Camera** and small window should appear.



Put a white paper to compensate for background effects. Background will not have effect if it is known in advance.
Please note: the USB model is not optimized for your camera or light conditions.

8. Press **q** to stop.

Create New Object

Task Description

The description of steps required to create a new model for a specific object is outlined below:

1. Install Pose6D setup and verify.
2. Select an object to work with- Lego blocks, shiny, small toy.
3. Camera connection and the setup- working distance, focus, object size in the image.
4. Camera Calibration
5. Video data acquisition of the selected object
6. Dimension measurements of the selected object
7. Transfer of the video data and dimensions to RobotAI.
8. RobotAI generates a model for the object.
9. Live Camera model testing on the selected object.

Pose6D Installation & Verification

Has been performed in the previous sections.

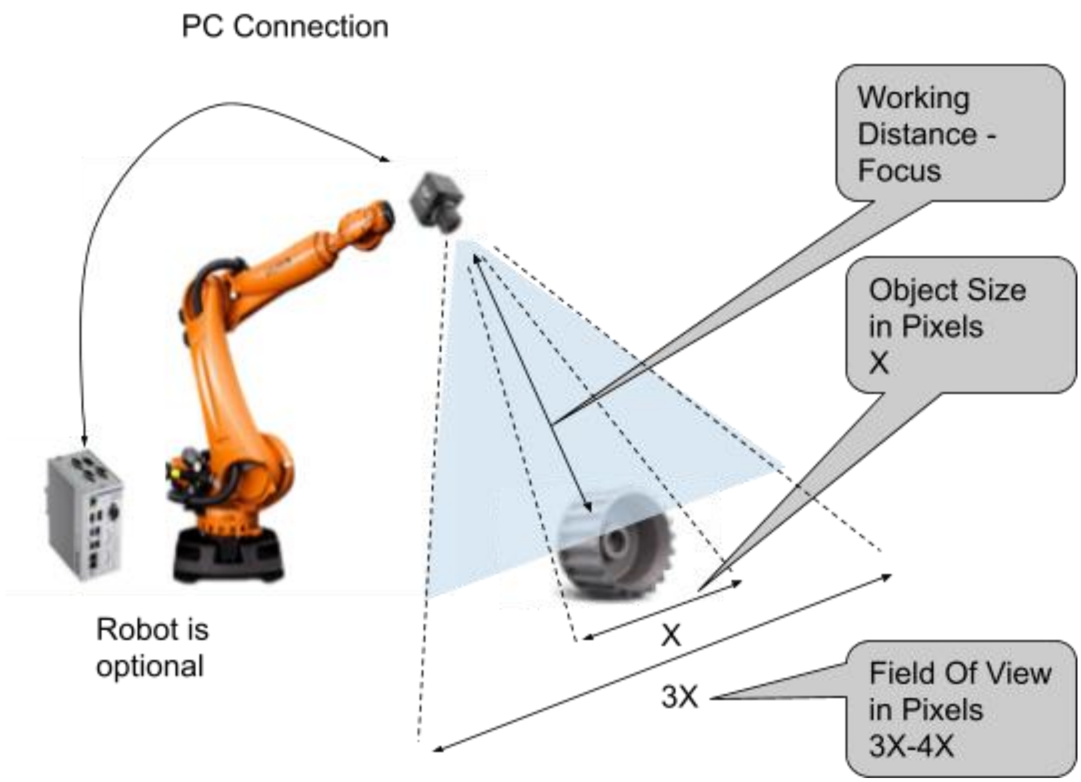
Object Selection

You may discuss with RobotAI how to start with a simple object.

Setup

The following is an example of the computer and the camera connection required for POC.

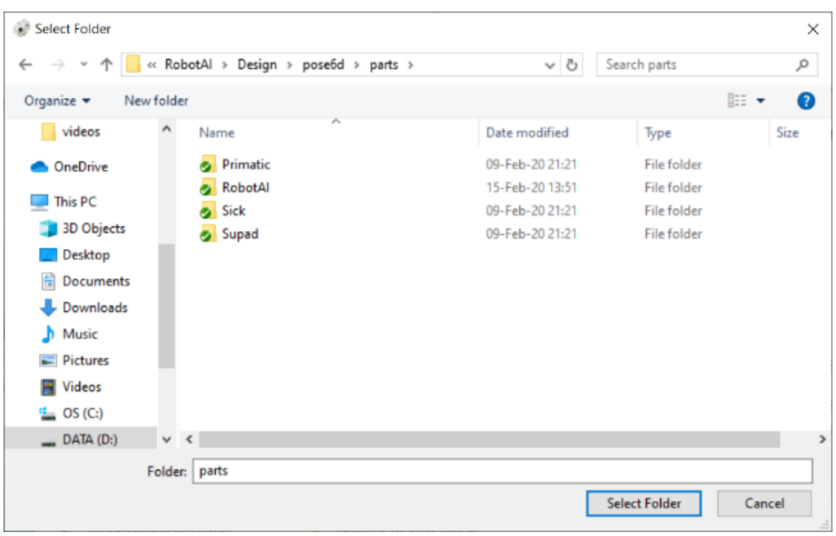
The main and the most important requirement is to have an object size at least 200x200 pixels, with image size around 1280x720 pixels. Selecting the camera and optics helps define the working distance.



The object size should be at least 200x200 pixels, which should occupy ⅓ or ¼ of the entire field of view.

In this step, a new directory is created for object position estimation.

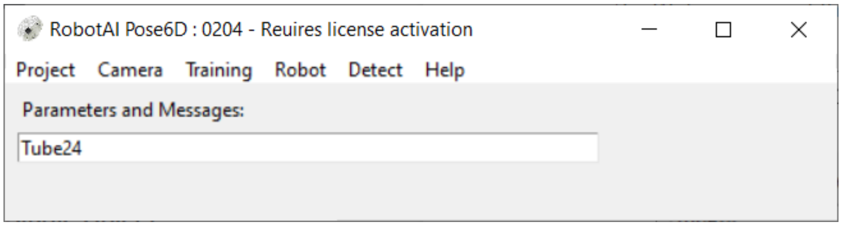
1. Set the working directory, which in our case will be **C:\RobotAI\Parts**, but can be



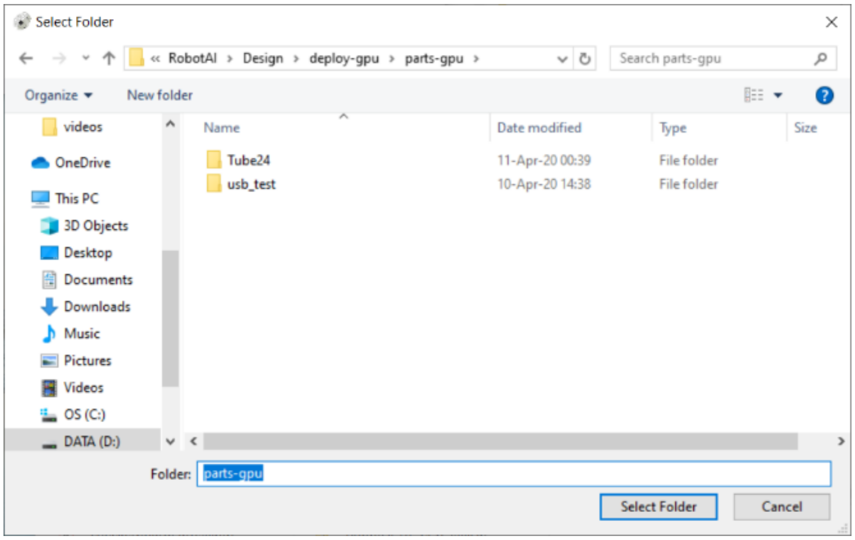
anything:

Menu bar> Project > Set Work Directory

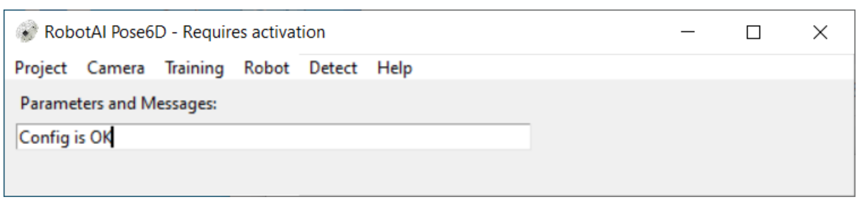
2. Write a name of the object you want to handle in the parameter box. In our case it is **Tube24**:



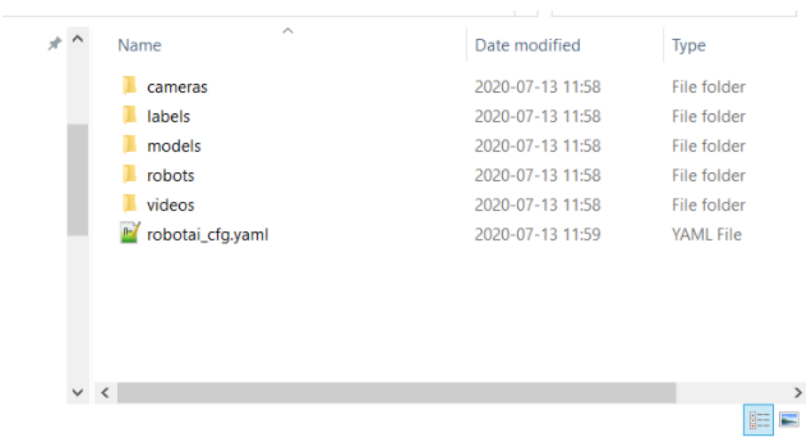
Then select: **Projects > Create New Object** to create a folder with this name:



3. Run **Menu bar> Projects > Check Config File>** to check that everything is created correctly (it should look like below):



There are 5 directories created under the folder Tube24:



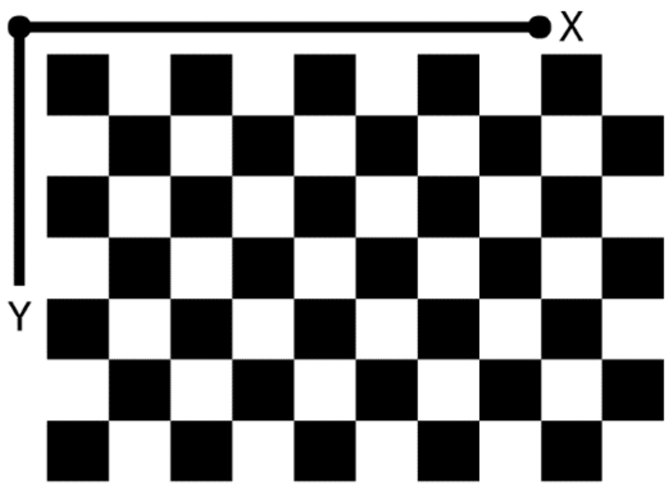
4. Recheck that you have an example json file created in the **labels** directory, which describes the object information, it should be edited for your specific object.

Camera Calibration

The following steps are used to estimate camera parameters in order to get correct measurements.

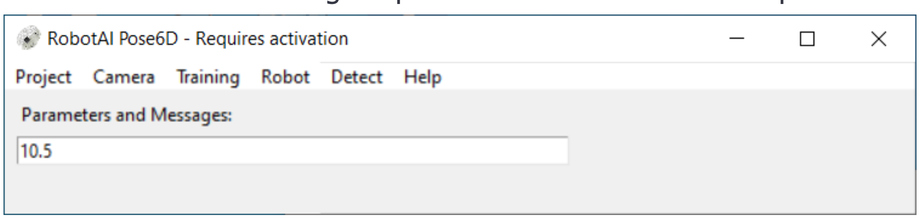
The following camera calibration procedure is required:

1. Select a USB camera to work with.
2. Connect to this camera and configure resolution and focus as described before. Keep them fixed.
3. Download the PDF files with chess pattern from [Chess Pattern](), this link[1]:



   Print it or use your mobile device/iPad to display it.

4. Measure the size of a single square in **mm** and set it in the parameter box:



   Click **Menu bar>Camera>Square Size>** to save it for the SW. The console window will indicate success:

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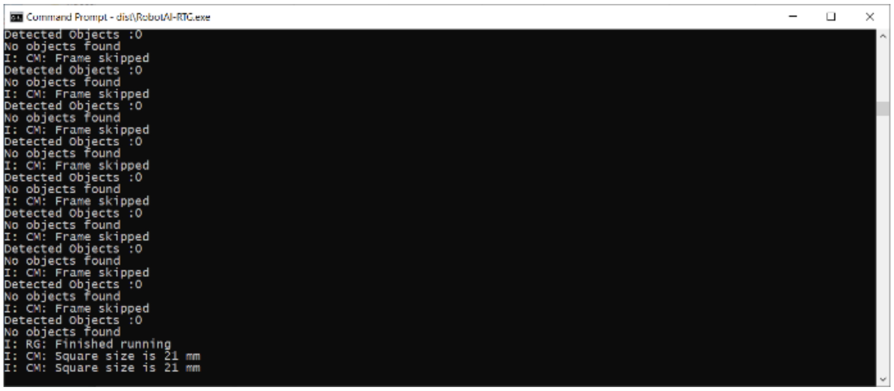
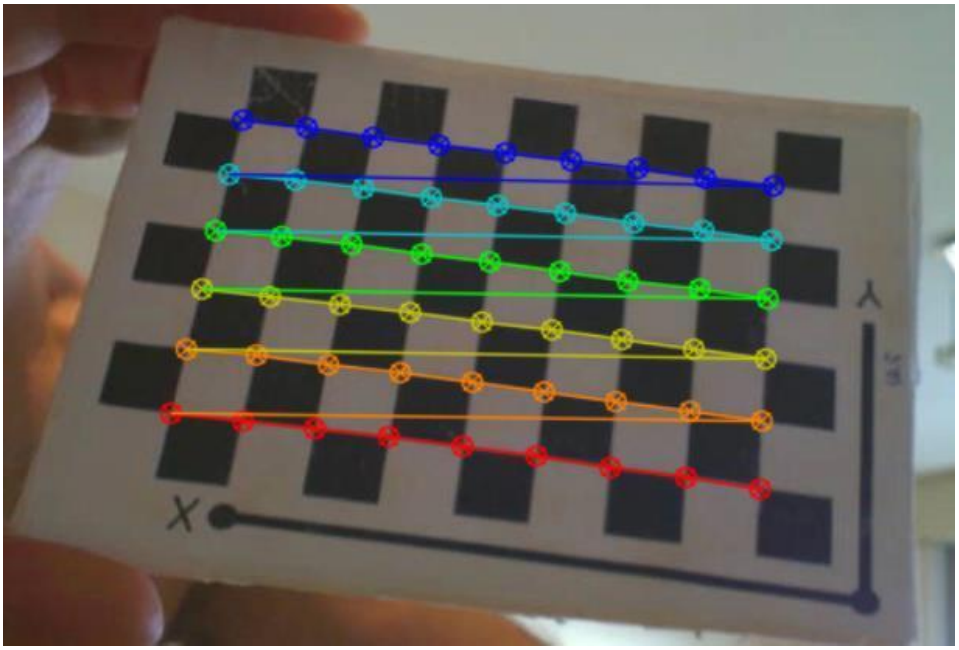
[1] https://drive.google.com/file/d/18VLDSXXj0uP_K44CRhA40ZHGqAcmvXMh/view?usp=sharing



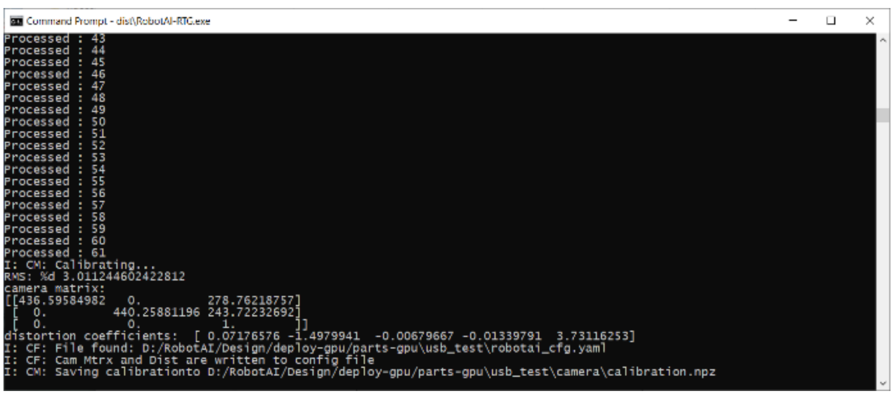
You can also see it in configuration file: **square_size** field.

5. After enabling the connection to your camera and setting the resolution, record a video for calibration by moving the camera or chess pattern relative to each other. The command is: **Menu bar> Camera > Record Video for Calibration**.
   You can watch an example by going to **C:\RobotAI\Parts \ usb_test \ cameras** and watching the mp4 file. This command will open a video stream from your camera and create an mp4 file.
6. There are several options available. Press the following keys when the image is shown: **q**-to stop, **n**-just show, **a**-acquire single image, **f**-acquire all, **t**-acquire timed, each 5 sec.
7. Around 20-30 images are enough. Check **calibration.mp4** in the **C:\RobotAI\Parts \Tube24\cameras>** directory.
8. You can create a calibration video several times, but you must remove the file manually. It is important to move the chess pattern relative to the camera, keeping it inside the video frame.

   Do not move it quickly. Do not smear the image.
9. Run: **Menu bar> Camera > Calibrate from File** and the software automatically selects and runs the calibration video files created by you. It should look like this:

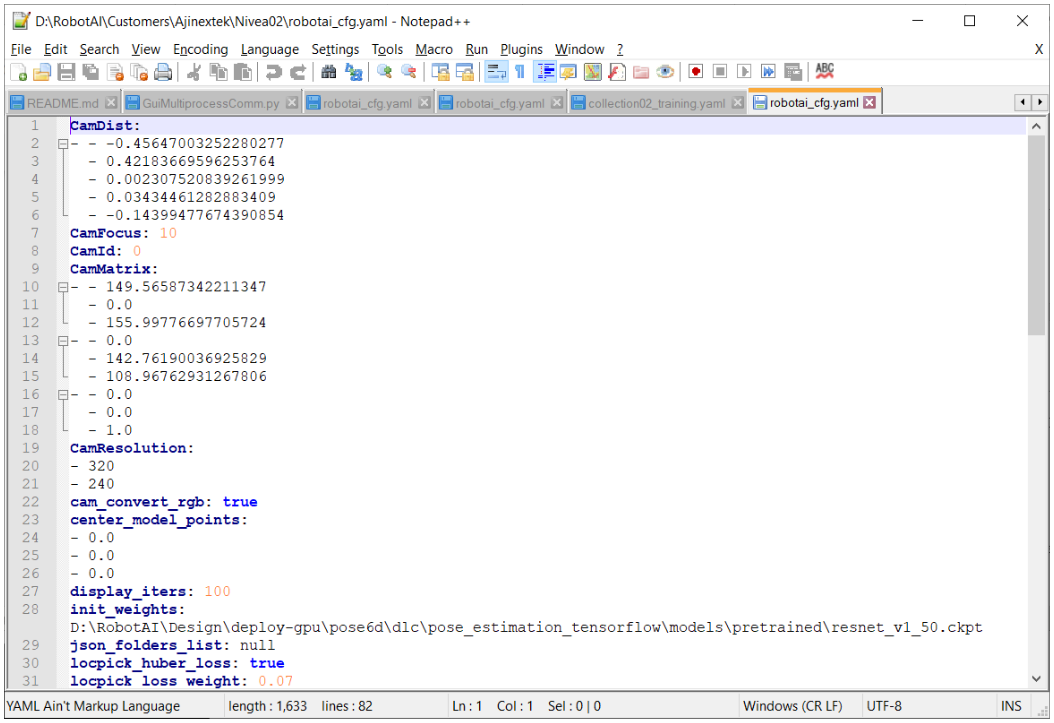


10. Press **q** at the end of the movie (if it is too long) to start the calibration sequence.
11. The console shows camera parameters and they are updated in **C:\RobotAI\Parts\Tube24\robotai_cfg.yaml**:



```
Command Prompt - dist\RobotAI-RTG.exe
Processed : 43
Processed : 44
Processed : 45
Processed : 46
Processed : 47
Processed : 48
Processed : 49
Processed : 50
Processed : 51
Processed : 52
Processed : 53
Processed : 54
Processed : 55
Processed : 56
Processed : 57
Processed : 58
Processed : 59
Processed : 60
Processed : 61
I: CM: Calibrating...
RMS: %d 3.011244602422812
camera matrix:
[[436.59584982   0.         278.76218757]
 [  0.         440.25881196 243.72232692]
 [  0.           0.           1.        ]]
distortion coefficients:  [ 0.07176576 -1.4979941  -0.00679667 -0.01339791  3.73116253]
I: CF: File found: D:/RobotAI/Design/deploy-gpu/parts-gpu\usb_test\robotai_cfg.yaml
I: CF: Cam Mtrx and Dist are written to config file
I: CM: Saving calibrationto D:/RobotAI/Design/deploy-gpu/parts-gpu\usb_test\camera\calibration.npz
```

12. The camera is calibrated for this specific resolution. Go to: **Menu bar>Project>Edit Config File** to open the config file and check that the calibrated parameters are written in the file:




```yaml
CamDist:
- - -0.45647003252280277
  - 0.42183669596253764
  - 0.002307520839261999
  - 0.03434461282883409
  - -0.14399477674390854
CamFocus: 10
CamId: 0
CamMatrix:
- - 149.56587342211347
  - 0.0
  - 155.99776697705724
- - 0.0
  - 142.76190036925829
  - 108.96762931267806
- - 0.0
  - 0.0
  - 1.0
CamResolution:
- 320
- 240
cam_convert_rgb: true
center_model_points:
- 0.0
- 0.0
- 0.0
display_iters: 100
init_weights:
D:\RobotAI\Design\deploy-gpu\pose6d\dlc\pose_estimation_tensorflow\models\pretrained\resnet_v1_50.ckpt
json_folders_list: null
locpick_huber_loss: true
locpick_loss_weight: 0.07
```


The calibration of the internal camera internal parameters is complete.

The calibration can be done even using your smartphone as shown here[2].

Training Data Acquisition

Perform relative motion of the camera or object, keeping the object inside the image with size of 200x200 pixels at least. Please check different examples and choose the suitable one. Moving camera link[3]. Fixed camera link[4].

Object Size Measurements

Short explanation of how to do measurements of the object to create a 3D model.

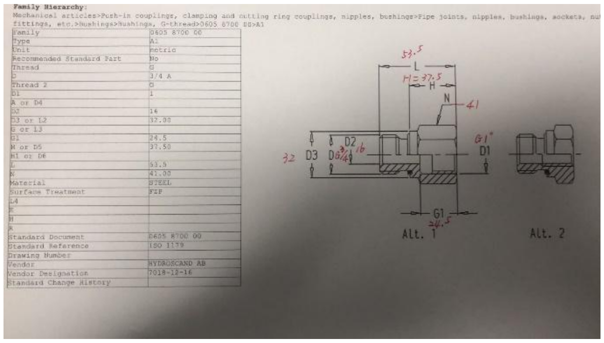
Option 1

Provide a mechanical drawing, like below.

[2] https://youtu.be/ORet1a4GR2o

[3] https://youtu.be/KSzGLQCJPtw

[4] https://youtu.be/XljXi0lNADE



Family Hierarchy:
Mechanical articles>Push-in couplings, clamping and cutting ring couplings, nipples, bushings>Pipe joints, nipples, bushings, sockets,
fittings, etc.>bushings>Bushings, G-thread>0605 8700 00>A1

Family	0605 8700 00
Type	A1
Unit	metric
Recommended Standard Part	No
Thread	G
D	3/4 A
Thread 2	G
D1	1
A or D4	
D2	16
D3 or L2	32.00
G or L3	
G1	28.5
H or D5	37.50
H1 or D6	
L	53.5
N	41.00
Material	STEEL
Surface Treatment	FZF
L4	
E	
H	
R	
Standard Document	0605 8700 00
Standard Reference	ISO 1179
Drawing Number	
Vendor	HYDROSCAND AB
Vendor Designation	2018-12-16
Standard Change History	

Option 2

Perform actual measurements, like this:

Sending Data to RobotAI and Creating the Model

Zip the object folder by pressing **Menu bar> Training > Zip Object Model.** This command will create a zip file with your model name. Send it to RobotAI.

RobotAI will create a model and send you back the model.

Unzip the model to the working folder like **C:\RobotAI\Parts \Tube24**.

Live Camera Testing the Model

After receiving zip file:

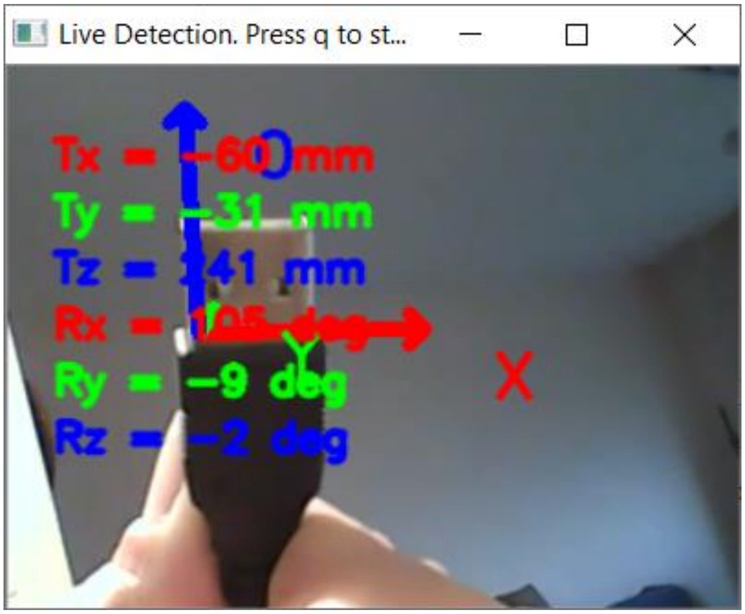
1. Download the model zip file to the directory **C:\RobotAI\Parts** .
2. The name of the zip should match the name of the object.
3. **Menu bar> Training > Unzip Model Zip File** will update the latest model and config file.
4. Perform steps similar to the '**usb_test**' model described previously: connect to the camera, set resolution.

5. Test the model on the previous data by selecting **Menu bar> Detect> Run Standalone** from File.
6. Connect to the camera and run: **Menu bar> Detect > Run Stand Alone with Camera**.

The following image should be shown:

If the software is used without license, only the 3D axis is shown.
To enable exact measurement output, the software requires licensing as described below.

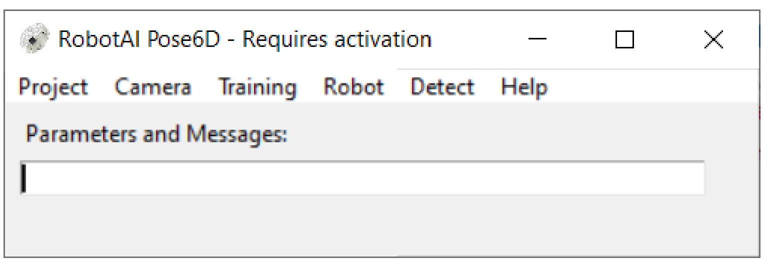
If the license is activated, a similar image to the below image is displayed:



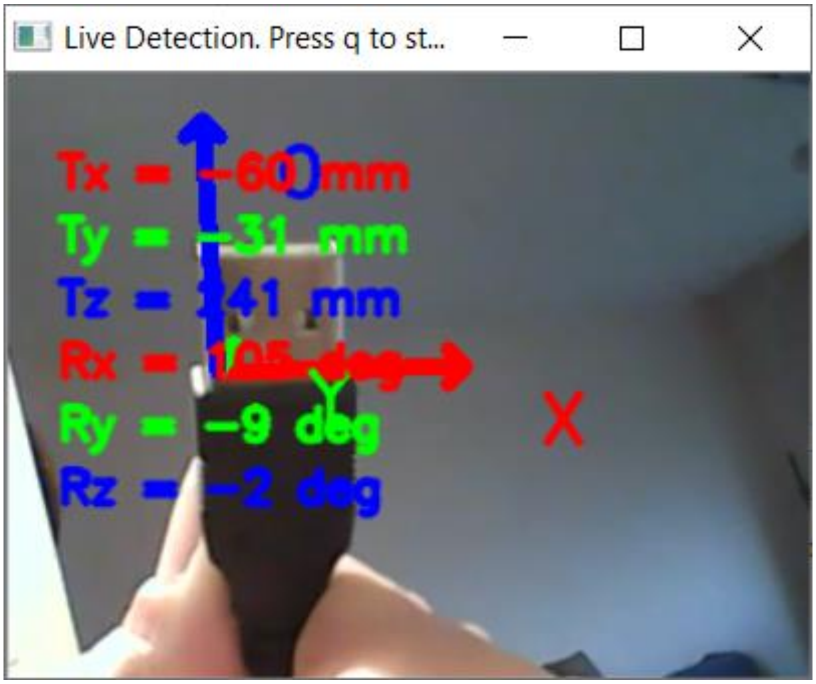
Software Activation & License Types

This section outlines the license activation procedure.

1. Running the software from directory '**C:\RobotAI\SW**' it will show the following string "Requires Activation"

   You will find license file "pose6d_license.chk" created in the directory '**C:\RobotAI\SW**'
2. Send this file by email to the RobotAI contact person along with the customer name.
3. You should receive email from RobotAI that will contain the similar file "pose6d_license.chk". This file will enable the software for specific machines, limited time period and certain features.
4. License types affect connection to the robots and the information displayed.
5. Testing with built in **usb_test** will show actual numbers estimated by SW:


Live Detection. Press q to st...
Tx = −60 mm
Ty = −31 mm
Ry = −9 deg
Rz = −2 deg
X
Y